MILITARY SPECIFICATION

MOUNT, PERISCOPE: M115

1. SCOPE

1.1 This specification covers one type of instrument designated Mount, Periscope: M115.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Acquisition, Disposition and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables: For Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F8620000	Mount, Periscope: Ml15
F8636535	Adapter
A8644448	Push-Pull Gage
F10554660	Holding Fixture
F11728871	Test Standard for Mount, Periscope: M115
F11728872	Set-Up Final Test Fixture for Mount,Periscope: M115

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PACKAGING DATA SHEET

8620000 Packaging of Mount, Periscope: M115

(Copies os specifications, standards, drawings and publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The mount shall be manufactured in accordance with Drawing F8620000, "Mount, Periscope: M115" and all drawings pertaining thereto.

3.3 General specification.- The contractor shall be responsible for adherence to, and compliance with the following requirements of MIL-F-13926:

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.4 Environmental.

3.4.1 Vibration. - The mount shall be vibrated in a vertical plane at a constant frequency of 30 cycles per second (CPS) with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes, plus or minus 15 seconds. Subsequent to vibration the mount shall show no evidence of loose parts or failure to operate. The headrest assembly shall not have changed position from it's original settings and the mount shall meet all other requirements of this specification.

3.4.2 Extreme operating temperatures.- The mount shall meet all the operability requirements of 3.6.1, 3.6.2, 3.6.3 and 3.6.4 while exposed and thermally stabilized at -40° and +150°F.

3.5 Orientation. - The mount shall include the shield, cover and headrest assembly. The locating surface of the mounting flange of the mount shall be secured in a horizontal plane as shown on Assembly Drawing F8620000.

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3.6 Performance.

3.6.1 Shield assembly.- With the plunger depressed (disengaged), maximum operation of the handle shall operate the shield through full operation; open for viewing and closed for protection. When the shield is fully open and when the shield is fully closed the plunger shall engage and lock the handle when the plunger is released.

3.6.2 Holding features. - A medium which incorporates the critical mounting surfaces of a Periscope M31 shall be utilized for insertion and removal on the mount to determine the following functional requirements.

3.6.2.1 Release grip. - When the medium is inserted into the periscope mount, the mount shall automatically latch the medium in the fully inserted position so that it cannot be removed without actuating the release grip. When applying an upward force of 15 to 20 pounds on the medium, a direct pull of not more than 12 pounds on the release grip shall actuate the grip to it's full disengaged position. Full diengagement of the grip shall allow removal of the medium from the periscope mount without restrictions. The recorded pull on the grip shall not change by more than 3 pounds when the mount is subjected to the temperatures specified in 3.4.2.

3.6.2.2 Lever. - When the plunger of the Level Assembly 8298434 is depressed and lever pushed to its fullest forward position and locked by releasing the plunger, the medium shall be secured in the mount. subsequent to the application of a 10 pound pressure applied horizontally to the front and rear of the medium at its lowest point, the medium shall be within .0005 inch of its original location when measured horizontally at that point. This requirement shall be met following the release of each of the pressures applied.

3.6.3 Headrest travel.

3.6.3.1 Headrest, horizontal travel. - Plunger 8262113 shall engage in the detent at each end of the slide and firmly secure the headrest in each position. When the plunger is freed from the detent, the headrest shall slide smoothly without interference when pushed right and left between the detent positions.

3.6.3.2 Headrest, rotational travel.- The headrest assembly shall provide full movement without interference when rotated through the maximum travel range permitted by Bracket 8229065, and shall not move when left in any position within the rotational travel range.

3.6.4 Adjustment range. - Each Screw Assembly B7639274 shall be loosened and the headrest assembly shall be adjustable through maximum movement permitted by the elongated slots in the assembly without restriction or force.

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3.6.5 Retention. - The torsional load applied to each bolt and wingnut that retains the headrest shall be such that the headrest shall not change position when subjected to the vibration specified in 3.4.1 and shall be operable at the temperatures specified in 3.4.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definition of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) mounts plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the

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contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and sub-assemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE - CLASSIFICATION OF DEFECTS

Class	Requirement	Test Procedure
Critical: None		
Major: AQL 1.0% Defective		
101. Vibration	3.4.1	4.6.1.1
102. Retention	3.6.5	4.6.1.1.1
103. Shield assembly	3.6.1	4.6.3.1
104. Release grip	3.6.2.1	4.6.3.2
105. Lever	3.6.2.2	4.6.3.3
106. Headrest, horizontal travel	3.6.3.1	4.6.3.4
107. Headrest, rotational travel	3.6.3.2	4.6.3.5
108. Adjustment range	3.6.4	4.6.3.6
Minor: None		

NOTE: The tests in Table I shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General. One mount shall be selected at random by a government representative as a special sample from each 100 produced, or from each months production, whichever occurs first. The sample shall have

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successfully met all acceptance tests specified herein prior to conducting the following examinations and tests.

TABLE II

No.	Characteristic	Requirement	Test Procedure
301.	Materials	3.1	Drawings & Specifications - Visual
302.	Fabrication	3.2	Drawings - Visual
303.	General specifications	3.3	MIL-F-13926, Visual

4.4.2 Environmental. - Three mounts shall be selected at random as a special sample from each 50 produced or from each months production, whichever occurs first. The samples shall meet the requirements and tests in Table 111 and shall then meet the requirements and tests in Table I.

TABLE III

No.	Characteristic	Requirement	Test Procedure
304.	Shield assembly	3.4.2, 3.6.1	4.6.1.2, 4.6.3.1
305.	Release grip	3.4.2, 3.6.2.1	4.6.1.2, 4.6.3.2
306.	Lever	3.4.2, 3.6.2.2	4.6.1.2, 4.6.3.3
307.	Headrest, horizontal travel	3.4.2, 3.6.3.1	4.6.1.2, 4.6.3.4
308.	Headrest, rotational travel	3.4.2, 3.6.3.2	4.6.1.2, 4.6.3.5
309.	Adjustment range	3.4.2, 3.6.4	4.6.1.2, 4.6.3.6

NOTE: The tests for Characteristics 304 through 309 inclusive, shall be conducted at -40° and +150°F.

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise porvided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-1-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the comtract, all inspection equipment specified by drawing number in specification

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or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP) l

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the “Test Methods and Procedures” specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and “Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error of test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Vibration.- Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.4.1 shall be utilized for this test. Unless otherwise specified, the test equipment shall be in accordance with the “Test Facilities” requirements of MIL-F-13926. The mount shall be secured to the vibrating equipment by means of an adapter that conforms to Drawing F8636535. The vibration motion shall be in the vertical plane. Prior to vibration the headrest assembly shall be positioned as shown on Drawing F8620000, Mount Periscope: M115. The headrest shall be located at its midpoint of travel in the elongated slots; each Screw Assembly 7639274 shall be hand-tightened; Plunger 8262113 shall be engaged in the detent at the right end of the slide; the shield shall be raised and the handle assembly set in the upper detent. Temporary reference marks shall be placed on the headrest assembly, bracket and plate assembly, etc. for the purpose of detecting any change in position of the headrest due to vibration. The mount shall then be vibrated for the time period specified in 3.4.1. Immediately following the vibration cycle, and prior to the examination for physical defects resulting from poor workmanship and materials, the mount shall be subjected to the test in 4.6.1.1.1. Subsequent to the test in 4.6.1.1.1, the mount shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.4.1 prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts prior to subjecting the mount to the remaining performance tests. Minimum visual examination shall include the shield assembly, the mount, the plate assembly, headrest assembly, nuts, etc. A tactile examination shall also be conducted to determine looseness or improper functioning of the above items as well as such moving parts

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as the latching mechanism, handle assembly, headrest assembly and lever assembly, etc. Any of the possible defects defined above shall be cause to consider a mount defective. Mounts which have passed the above examination shall be subjected to all other tests of this specification.

4.6.1.1.1 Tetention. - Immediately following vibration, the shield and the temporary reference marks shall be subjected to a visual examination. The shield shall not have closed and the temporary reference marks shall indicate no movement or change in position of the headrest due to vibration.

4.6.1.2 Extreme operating temperatures.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.2. The mount shall be placed in the chamber and the temperature lowered gradually until the chamber temperature is -40°F. The mount shall be kept at this temperature for a period of two hours. While still at the specified lower temperature the mount shall then be subjected to the tests specified in 4.6.3. Upon completion of these tests the mount shall be returned to standard ambient temperature and it shall be visually examined for any failures in workmanship or materials. The mount shall again be placed in the test chamber and the temperature gradually increased until the chamber temperature is 150°F. The mount shall be kept at this temperature for a period of 2 hours. At the end of this period and while still at the specified higher temperature the mount shall be subjected to the tests specified in 4.6.3. When the above tests have been completed the mount shall be returned to standard ambient temperature and once again visually examined for any failures in workmanship or material.

4.6.2 Orientation.- Special test equipment conforming to the requirements of Drawing F10554660 (Holding Fixture) shall be utilized to accept the mount, completely assembled, and secure it in the position required by 3.5. Set-up of the test equipment and installation of the mount under test shall be performed in accordance with the applicable set-up instructions contained on Drawing F11728872 (Set-Up Final Test Fixture for Mount Periscope: M115).

4.6.3 Performance. - To determine compliance with 3.6, the following tests shall be performed with the complete mount positioned on the special test equipment specified in 4.6.2.

4.6.3.1 Shield assembly.- To determine compliance with 3.6.1, the locking plunger shall be released from its detent and the handle raised to open the protective shield. The plunger shall then be positioned in the upper detent where it shall lock the shield in the opened position. The plunger shall be depressed once again and the handle lowered to close the shield." When the plunger is engaged in the lower detent the shield shall be locked in the closed position. The clearance between the shield and the mount shall be 1/16 ± 1/32 inch as shown in Drawing F8620000, Mount, Periscope: M115.

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4.6.3.2 Pelease grip. - The test standard conforming to the requirements of Drawing F11728871 shall be utilized for this test. Rotate the headrest to its uppermost position. Release the Lever Assembly 8298434 and pull to full open position. The headrest ad the lever assembly shall remain at these positions during this test. Insert the test standard into the bottom of the mount and push upward until the release grip (Safety latch) is activated and snaps into position. The release grip shall hold the test standard in the mount as required by 3.6.2.1. Caution shall be exercised before releasing or removing hands from test standard to prevent injury, if the release grip fails. Upon satisfactory completion of this portion of the test, an upward force, within the range specified in 3.6.2.1, shall be applied to the test standard. With this upward force applied, actuate the release grip to its fully disengaged position utilizing the push-pull gage conforming to the requirements of Drawing A8644448. The direct pull required to actuate the release grip to its fully disengaged position shall not exceed the applicable values specified in 3.6.2.1 for the temperatures indicated. Full disengagement of the release grip shall allow the test standard to be removed freely from the mount without any restrictions (hang-up) as specified in 3.6.2.1. At no time shall the test standard be allowed to drop freely and strike hard against any solid object.

4.6.3.3 Lever. - With the Test Standard 11728871 inserted as in 4.6.3.2, release the Lever Assembly 8298434 and push to its forwardmost position. This action will lock the test standard into the mount and the spring loaded plunger will secure the lever assembly in a locked position. A dial indicator graduated in .0001 inch increments shall be positioned so that it contacts the front or the rear button provided on the left side of the test standard, and then adjusted to read zero with respect to the indicating hand. Apply a slow steady pressure of 10 pounds, by means of the pus-pull gage specified in 4.6.3.2, to the opposite side of the dial indicator contact point; and then upon releasing the pressure, the indicating hand of the dial indicator shall not exceed the tolerance specified in 3.6.2.2. Reverse the equipment and repeat the above procedure. Upon satisfactory completion of this portion of the test, repeat the entire above procedure at the front and rear buttons on the right side of the test standard.

4.6.3.4 Headrest, horizontal travel.- push the headrest until the Plunger 8262113 engages the detent at each end of the slide. The headrest shall be firmly secured in each position when subjected to a tactile examination. While the plunger is disengaged from the detent, the headrest shall be pushed right and left (a minimum of 3 times) between the detent positions. The headrest shall slide smoothly without interference and evidence of binding.

4.6.3.5 Headrest, rotational travel. - Rotate the headrest assembly through its maximum rotational travel range a minimum of 4 times. During these rotations, a tactile examination shall be made for undue irregularities of movement and looseness. Upon successful completion of this portion of the test, rotate the headrest assembly to its midpoint of travel. When left in this position, the headrest assembly shall not move. In addition to checking the headrest assembly at this position, repeat this procedure

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at a point midway between the midpoint and each extreme position and also at each extreme position for a total of 5 checks overall.

4.6.3.6 Adjustment range.- When each Screw Assembly 7639274 is loosened in the headrest assembly, the Headrest 8267966 (with Frame 8620018) shall be translated front to rear in the elongated slots of Bracket 8229065. During this translation, there shall be no restriction to the movement of the headrest assembly as specified in 3.6.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking shall be in accordance with Packaging Data Sheet 8620000. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Mount, Periscope: M115 is primarily intended for use in the fire control system of the 105MM Gun, Tank, M60 Series and is used to support the Periscope M31.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of preservation, packaging and packing.
c. Applicable packaging data sheet (see 5.1).
d. Applicable stock number.
e. Provisions for first article testing.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A377

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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